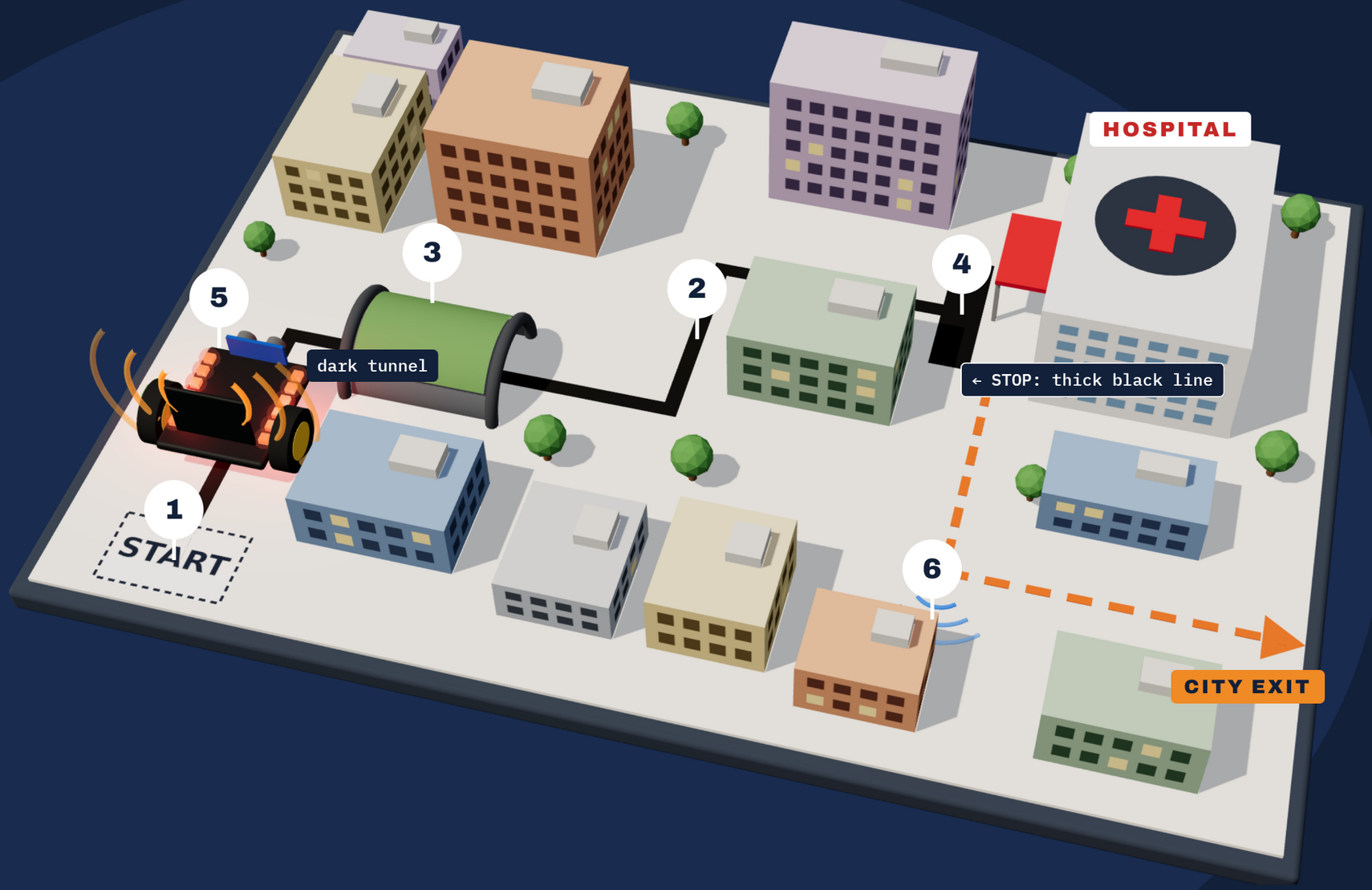
GCSE COMPUTER SCIENCE · MICRO:BIT + BIT:BOT · PYTHON

MISSION: AMBULANCE RUN

Your Bit:Bot is an ambulance. Get the patient through the city to the hospital, then find a safe way back out without hitting a single building.



BITBOTZERO TOOLKIT

speed 0–100 · LEDs 0–11 (0–5 left arm, 6–11 right) · colours: bot.RED, GREEN, BLUE, YELLOW, PURPLE, WHITE

SET UP

```
from microbit import *
import bitbotXL as bot
```

Bit:Bot Classic? Import bitbotClassic instead.

MOVE

```
bot.move_forward(speed)
bot.move_backwards(speed)
bot.turn_left(speed)
bot.turn_right(speed)
bot.stop()
```

add ms to move for a set time

LIGHTS + SOUND

```
bot.led_on(n, bot.WHITE)
bot.led_off(n)
bot.leds_on(bot.RED)
bot.leds_off()
bot.buzzer(ms)
```

buzzer waits until the beep ends

SENSE

```
bot.get_left_line()
bot.get_right_line()
```

1 = over the line, 0 = not

```
bot.get_left_light()
```

0 = dark ... 1023 = bright

```
bot.get_distance()
```

cm, or -1 if no echo

MISSION STAGES

1 Set off — button B

Wait on the start pad. Only drive once B is pressed.

```
button_b.was_pressed()
bot.move_forward(30)
```

Think: what should the robot do while it waits?

2 Follow the line — 2 × infra-red sensors

Steer between the buildings using the left and right line sensors.

```
bot.get_left_line()
bot.turn_left(speed)
```

Think: only the left sensor sees black. Which way do you turn?

3 Tunnel — light sensors

When it gets too dark, switch on the first two lights.

```
bot.get_left_light()
bot.led_on(0, bot.WHITE)
```

Think: `print()` the light level in and out of the tunnel. Where's your threshold?

4 Hospital arrival — both sensors on black

At the thick black line, stop and run an arrival routine: sound and lights.

```
bot.stop()
bot.buzzer(500)
bot.leds_on(bot.GREEN)
```

Think: why must *both* sensors see black? How do you stop it triggering twice?

5 Emergency mode — button A · any time

Press A: siren sounds and lights flash red. Press A again: back to normal.

```
button_a.was_pressed()
bot.leds_on(bot.RED)
bot.buzzer(100)
```

Think: how will your program remember it's in emergency mode? Can it still steer while it beeps?

6 Route out — ultrasonic sensor

Take a different route out of the city. Building too close? Turn. Never crash.

```
bot.get_distance()
bot.turn_right(speed, ms)
```

Think: it returns -1 if there's no echo. What happens if you don't check for that? How many cm is "too close"?